

HARSH RAJ

 github.com/harshraj172  harsh777111raj@gmail.com  [linkedin.com/in/harsh-raj](https://www.linkedin.com/in/harsh-raj)  [Google Scholar](#)

EDUCATION

Northeastern University, Boston, MA
Master of Science in Computer Science

May 2027 (expected)
GPA: 3.6/4.0

Delhi Technological University, New Delhi, India
Bachelor of Technology in Engineering Physics

August 2023
GPA: 3.4/4.0

TECHNICAL SKILLS

Languages: Python, C++, CUDA, Java, SQL, HTML, JavaScript, Bash

Frameworks/Libraries: PyTorch, FastAPI, RayServe, RabbitMQ, MongoDB, DynamoDB, Redis, PyTest, VectorDBs

DevOps/Cloud: Docker, Kubernetes, Terraform, Azure, AWS, RunPod, Petals, CI/CD

Other: Git, Vim, SSH, Linux, RESTful APIs, Web-Sockets

Model Expertise: LLM Pre-training, LLM Post-training, Large Scale Data Synthesis

PROFESSIONAL EXPERIENCE

Research Assistant, Forschungszentrum Jülich
Data Curation and Scaling Laws

July 2025 – Present

- Contribute to EU Foundational Model projects ([OpenEuroLLM](#), [ELLIOT](#)) on JUPITER supercomputer
- Lead ablation studies on dataset mixtures and scaling laws on LLM training data at the petabyte scale
- **Impact:** Optimized multi-node data processing pipelines using Ray, achieving **5x** speedup in deduplication, filtering and map reduction operations

Co-Founder, Ontocord AI (Open Source)
Pre-training Data Curation

Jan 2024 – Jan 2025

- Co-founded open-source initiative focused on creating high-quality permissive datasets for LLMs
- Developed data processing pipelines with strategies for mixing, formatting, upsampling & interleaving
- **Impact:** Published [MixtureVitae-300BT](#) achieving **10%** performance improvement over existing permissive data sets

Applied Scientist, Vijil AI
LLM Red-Teaming

Feb 2024 – Nov 2024

- Contributed to the [vijil-evaluation](#) platform for red-teaming of enterprise AI agent workflows. Integrated more than **10** benchmark datasets for vulnerability assessment
- Developed the vijil-fuzzer for adversarial text mutation. It achieved a **85%** attack rate against GPT-4o
- **Impact:** The vijil-evaluate service is used by enterprise clients including [DigitalOcean](#) and [Groq](#)

Machine Learning Engineer, Yield Inc
Post-training and Data Curation

Sept 2022 – Jan 2024

- Built synthetic data generation and finetuning infrastructure for LLM agents - [synchaev](#) and [mandrill](#)
- Contributed to [Cacti](#), a Web3 transaction chatbot that simplifies blockchain interactions through natural language
- **Impact:** It was the first open-source chatbot that used custom tool calling (we called it [widgets](#)) before any LLM had native tool calling capabilities

SELECTED PUBLICATIONS

Improving Consistency in Large Language Models through Chain of Guidance **1st Author**

TMLR 2025

Defences against Reverse Preference Attacks **3rd Author**

AAAI 2025

Measuring Reliability of Large Language Models through Semantic Consistency **1st Author**, *Best Paper* NeurIPS 2022

PROJECTS

terminal-bench (800+ GitHub ★) Language agent sandbox evaluation framework (**Co-author & Maintainer**)

inspect-ai (1300+ GitHub ★) Industry-standard LLM evaluation suite (**Contributor**)

AWARDS AND ACHIEVEMENTS

[SWISS AI](#) Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding

2025

[Vision and Language Navigation](#) – Secured 3rd position (at the time of submission) on the R2R benchmark

2024

[Rank 7](#) – Codeforces Raif ML Round 1 Competition

2021